

CMOS 4K-BIT STATIC RAM

DESCRIPTION

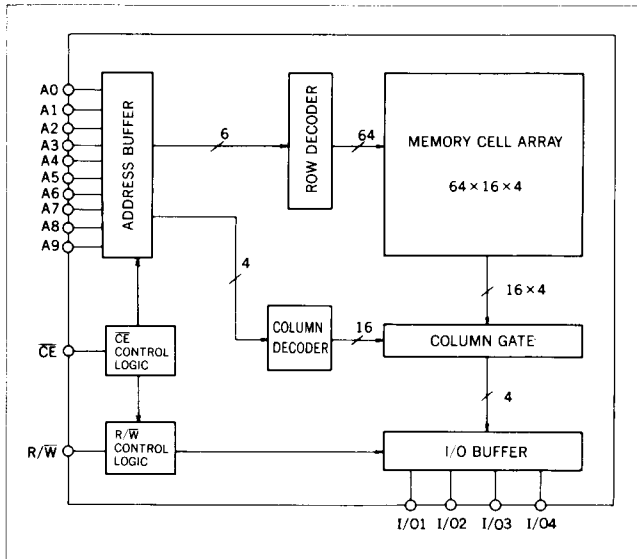
The SRM2114₂₅ is a 1,024 words x 4 bits asynchronous, static, random access memory on a monolithic CMOS chip. Its very low standby current consumption makes it ideal for applications requiring non-volatile storage with back-up batteries. The static nature of the memory requires no external clock or refreshing circuit.

Both the input and output ports are TTL compatible; and the three-state output allows easy expansion of memory capacity. These features make the SRM2114₂₅ usable for a wide range of applications from microprocessor systems to terminal devices.

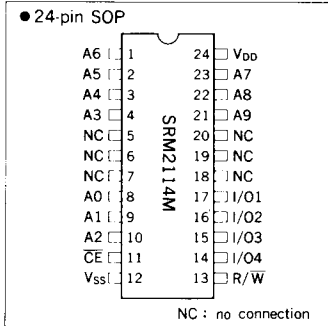
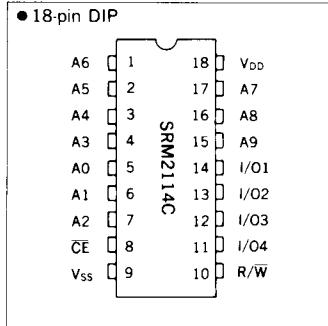
FEATURES

- Access time250ns (Max)
- Single power supply ...5V±10%/3 to 5V
- Low supply currentStandby : 0.1μA (Typ)/0.1μA (Typ)
Operation: 14mA(Typ)/9mA (Typ)
- Completely static operation
- TTL compatible inputs and outputs
- 3-state output with wired-OR capability
- Operation with back-up batteries
- Package 18-pin DIP(plastic)
24-pin SOP (plastic)

BLOCK DIAGRAM



PIN CONFIGURATION



PIN DESCRIPTION

A0 to A9	Address Input
R/W	Read/Write
CE	Chip Enable
I/O1 to 4	Data Input/Output
Vdd	Power Supply (+)
Vss	Power Supply (-)

ABSOLUTE MAXIMUM RATINGS

(V_{SS}=0V)

Parameter	Symbol	Ratings	Unit
Supply voltage	V _{DD}	-0.5 to 7.0	V
Input voltage	V _I	-0.5 to 7.0	V
I/O voltage	V _{IO}	-0.5 to V _{DD} +0.3	V
Power dissipation	P _D	1.0	W
Operating temperature	T _{opr}	-40 to 85	°C
Storage temperature	T _{stg}	-65 to 150	°C
Soldering temperature and time	T _{sol}	260°C, 10s (at lead)	—

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	SRM211425			Unit
		Min	Typ	Max	
Supply Voltage	V _{DD}	4.5	5.0	5.5	V
	V _{SS}	—	0	—	
Input Voltage	V _{IH}	2.2	—	V _{DD} + 0.3	V
	V _{IL}	-0.3	—	0.8	V

ELECTRICAL CHARACTERISTICS

DC Electrical Characteristics

(SRM211425 . . . V_{DD} = 5V ± 10%, V_{SS} = 0V, T_a = -40 to 85°C)

Parameter	Symbol	SRM211425				Unit
		Conditions	Min	Typ	Max	
Input leakage current	I _{LI}	V _I = 0 to V _{DD}	-1.0	—	1.0	μA
Standby supply current	I _{DDS}	C _E = V _{DD} - 0.2V	—	0.1	5	μA
Operating supply current	I _{DDO}	C _E = V _{IL} *C _L = 100pF	—	*14	*35	mA
Output leakage current	I _{LO}	V _O = 0 to V _{DD}	- 1.0	—	1.0	μA
High level output voltage	V _{OH}	I _{OH} = - 1.0mA	2.4	—	—	V
Low level output voltage	V _{OL}	I _{OL} = 2.0 mA	—	—	0.4	V

*C_L = 0pF Typ7 Max 25

Terminal Capacitance

(f = 1MHz, SRM211425 . . . V_{DD} = 5V ± 10%, V_{SS} = 0V, T_a = -40 to 85°C)

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Input capacitance	C _I	V _I = 0V	—	—	7	pF
I/O capacitance	C _{IO}	V _{IO} = 0V	—	—	10	pF

AC Electrical Characteristics

Read Cycle

(SRM211425 . . . V_{DD} = 5V ± 10%, V_{SS} = 0V, T_a = -40 to 85°C)

Parameter	Symbol	Conditions	SRM211425		Unit
			Min	Max	
Read cycle time	t _{RC}	*1	250	—	ns
Address access time	t _{ACC}		—	250	ns
C _E access time	t _{ACE}		—	250	ns
C _E output set time	t _{CLZ}	*2	0	—	ns
C _E output floating	t _{CHZ}		—	120	ns
Output hold time	t _{OH}	*1	10	—	ns

O Write Cycle

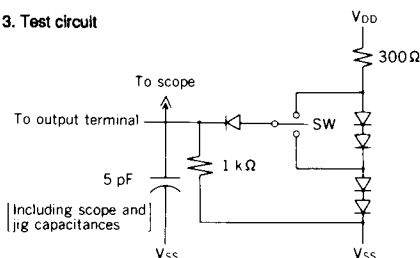
Parameter	Symbol	Conditions	SRM211425		Unit
			Min	Max	
Write cycle time	tWC	*1	250	—	ns
Address setup time	tAS		0	—	ns
Write pulse width	tWP		150	—	ns
Address hold time	tWR		0	—	ns
Input data setup time	tDW		100	—	ns
Input data hold time	tDH		0	—	ns
R/W output floating	tWHZ	*3	—	120	ns

***1 Test Conditions.**

1. Input pulse level : 0.8 to 2.2V
2. $t_r = t_f = 10\text{ns}$
3. Output load : $t_{HL} + C_L = 100\text{pF}$ (including scope and jig capacitances)
4. Timing reference level : 1.5V

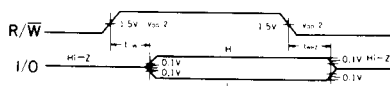
*3 Test Conditions

1. Input pulse level : 0.8 to 2.2V
2. $t_r = t_f = 10\text{ns}$
3. Test circuit



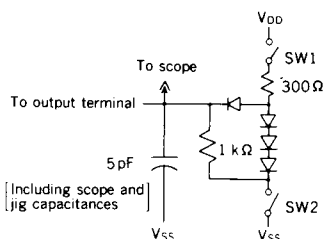
- SW is set to the V_{DD} side when measuring Hi-Z→"High" and "High"→Hi-Z of t_{OW} or t_{WHZ} .
- SW is set to the V_{SS} side when measuring Hi-Z→"Low" and "Low"→Hi-Z of t_{OW} or t_{WHZ} .

Output turn-on turn-off times



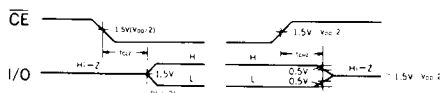
*2 Test Conditions

1. Input pulse level : 0.8 to 2.2V
2. $t_r = t_f = 10\text{ns}$
3. Test circuit



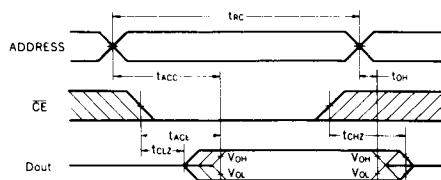
- SW1, SW2 are closed when measuring t_{CHZ} .
- SW1 is open, SW2 is closed when measuring $H_i-Z \rightarrow$ "High" of t_{CLZ} .
- SW1 is closed, SW2 is open when measuring $H_i-Z \rightarrow$ "Low" of t_{CLZ} .

Output turn-on turn-off times

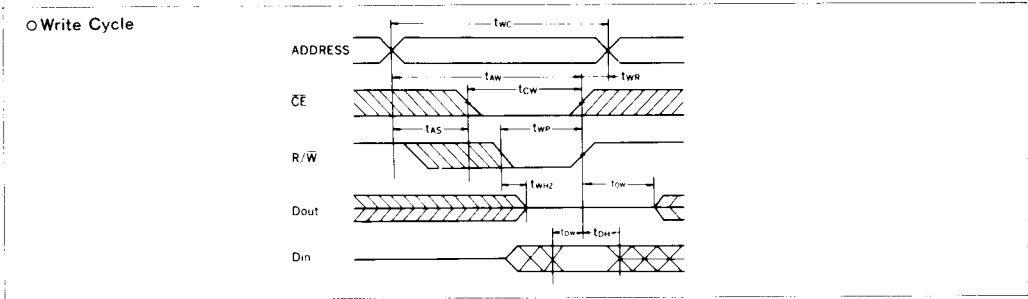


●Timing Chart

- Read Cycle



* $R/\overline{W}$ is "High" during read cycle.

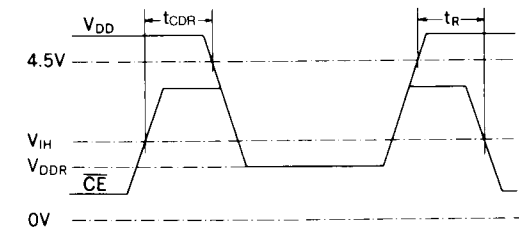


■ DATA RETENTION CHARACTERISTICS WITH LOW SUPPLY VOLTAGE

(SRM211425 Ta = -40 to 85°C)

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Data retention supply voltage	V _{DDR}	V _I = 0 or V _{DD} , $\overline{CE}$ = V _{DD}	2.0	—	5.5	V
Data retention supply current	I _{DDR}	V _{DD} = 2.0V, 25°C	—	—	2	μA
$\overline{CE}$ setup time	t _{CDR}	Refer to the Figure Below	0	—	—	ns
$\overline{CE}$ recovery time	t _R		t _{RC} *	—	—	ns

Data retention timing



* When retaining data in standby mode, supply voltage can be lowered within a certain range. But read or write cycle cannot be performed while the supply voltage is low.

FUNCTIONS

● Truth Table

$\overline{CE}$	R/ $\overline{W}$	A0 to A9	DATA I/O	MODE
H	X	X	Hi-Z	Unselected
L	L	Stable	Input data	Write
L	H	Stable	Output data	Read

X: "H" or "L"

● Reading Data

Data can be read out if an address is set while $\overline{CE}$ is "Low", and R/ $\overline{W}$ is "High".

● Writing Data

There are the following two ways of writing data into the memory.

(1) Hold $\overline{CE}$ "L", set the address, and apply low level pulse to R/ $\overline{W}$.

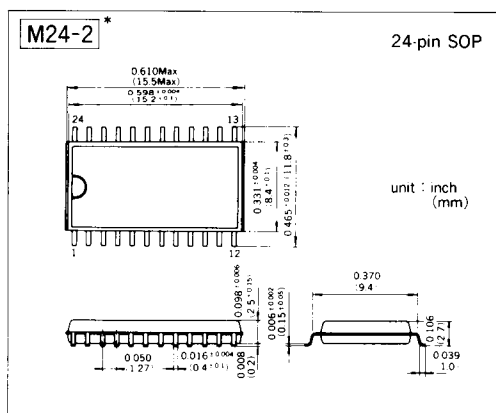
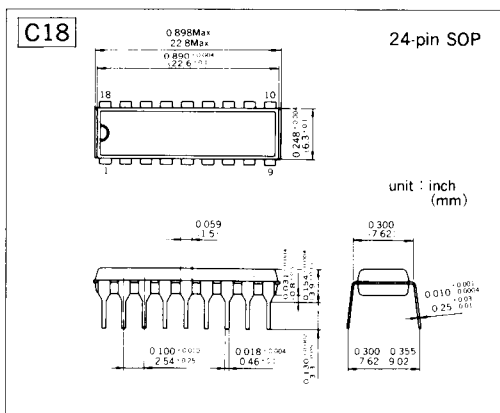
(2) Set the address and apply Low level pulse to $\overline{CE}$ and R/ $\overline{W}$.

In each case, data from the DATA I/O terminal is latched at the positive transition of "L" pulse of $\overline{CE}$ or R/ $\overline{W}$. The DATA I/O terminal is in high-impedance state when $\overline{CE}$ is "H" or R/ $\overline{W}$ is "L", so competition of data driver and memory output can be avoided.

● Standby Mode

When $\overline{CE}$ is "H" SRM211425 is in standby mode and only retains the data. In this mode, DATA I/O terminals are in high-impedance state, and all inputs of addresses, R/ $\overline{W}$, or data can be any "H" or "L". The current flowing in the SRM211425 is only leakage current.

PACKAGE DIMENSIONS



*Represents model SRM2114M25/L7 that has the same electrical characteristics as model SRM2114C25/L7.

■ CHARACTERISTICS CURVES

